

Ethan Wilson

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PROFESSIONAL SUMMARY

Applied Mathematics graduate with hands-on experience in data engineering, backend systems, and frontend development. Skilled in Python, Excel, SQL, React, and technical modeling, with a background in energy analytics. Seeking a software or data engineering role where I can contribute to technical products, improve data pipelines, and keep growing as a developer.

PROFESSIONAL EXPERIENCE

Data Analyst - Renewable Energy, Energeia - Davis, CA

Jun 2024 - Present

- Led data validation/preprocessing for the Moreno Valley Utility Integrated Resource Plan (IRP), performing weather normalization, interpolation, time-series forecasting, and regression-based modeling to generate 8760 hourly load profiles using Python and Excel. Supported final IRP reporting aligned with California Energy Commission (CEC) guidelines.
- Conducted in-depth research and quantitative modeling for San Jose Clean Energy's Building Electrification Workforce Assessment, analyzing contractor demographics and training programs to inform comprehensive outreach strategies.
- Utilized SQL (Snowflake), Python, and Energeia's uSim platform for data handling, validation, and weather normalization in support of Pepco Holdings' DER adoption analysis — modeling load and hosting capacity across 2,000 feeders and 300 substations across five operating companies in multiple jurisdictions to inform utility system planning.
- Designed advanced Excel models, incorporating macros and dynamic dashboards, to support the Colorado Energy Office's building electrification and decarbonization pathway optimization across seven state-level scenarios.
- Supported Tulare County's ZEV Study by analyzing utility load and GIS data to forecast peak demand and identify grid-constrained and underserved feeders and charging zones.

Frontend Developer Intern, Frontend Simplified - (Remote), CA

Jan 2024 - Jun 2024

- Enhanced a static single-page application built with HTML, CSS, JavaScript, and React by incorporating interactive features such as animations, transitions, and carousels to elevate the user interface.
- Leveraged Axios to handle API requests, dynamically fetched data from a cloud server, and implemented features like pagination, skeleton loading states, and dynamic routing for a seamless user experience.
- Utilized Git version control and the GitHub interface to work and thrive in a virtual and collaborative team environment
- This experience led to additional freelance opportunities in web development

Backend Software Engineer (Data Analytics Intern), Bluescape - Redwood City, CA

Jun 2023 - Sept 2023

- Worked on project focused on optimizing the deployment and management of microservices using Kubernetes and Helm for streamlined package management and Bash scripting for automation
- Leveraged Grafana and Loki to analyze company logs, utilizing regex and Linux commands to identify query errors
- Directly fixed Grafana dashboards, improving data representation and analytics tools for various engineering teams
- Acquired hands on experience in a wide range of technologies including AWS, Azure, Microsoft Office 365, Grafana, Loki, Prometheus, Python VS Code, Kubernetes, GitHub, LogCli, Linux, Atlassian tools, and SQL

EDUCATION

University of California, Davis - Applied Mathematics, B.S.

Aug 2021 - Dec 2023

- Relevant Courses:** ODE's, Real, Numerical, and Complex Analysis series, O.O.P. in C++, Minds, Brains, and Computers
- Activities and Extracurriculars:** Camp Kesem Counselor, Aquatic Specialist, Sigma Nu Fraternity, Rock Climbing and Chess Club Team, AI Student Collective, Mathematics Tutor

TECHNICAL SKILLS

- Languages:** Python, JavaScript, TypeScript, Java, C++, Swift (iOS), HTML5, CSS3 (BEM)
- Frameworks & Libraries:** React (with Hooks), Next.js, Redux, TensorFlow 2.0
- Data & Analysis Tools:** Excel, Snowflake (SQL), R, Jupyter Notebook, MatLab
- DevOps & Infrastructure:** Kubernetes, Helm, Grafana, Prometheus, AWS (S3, DynamoDB), Microsoft Office 365
- Version Control & Scripting:** Git, Bash Shell Scripting, Linux, Bitbucket, NPM/Yarn, Firebase v9
- Project Management & Methodologies:** Jira, Confluence, Wrike, Agile Methodologies
- Mathematics:** Calculus of Variations, Statistics/Probability, Set Theory, Logic, Analysis Series, Number Theory, NLPs

PROJECTS AND TECHNICAL EXPERIENCE

- Stock Market Prediction AI:** Used data analysis techniques and web scraping to collect stock data and train a sequential model using linear regression, neural networks, NLP, and data preprocessing
- Frontend Development:** NFT Online Market, E-Commerce Library, and Movie Search Database projects
- Vessel Traffic System:** Application design project on MATLAB utilizing a local database of ships and connected with ThinkSpeak, an IoT analytics platform, UI focused
- CSR Summer Programming Camp at Stanford University:** IOS applications using Swift, designed messaging app
- Log Analysis:** Automated large-scale log scanning in Loki using regex and shell scripts to extract and flag error patterns.
- Arduino Engineering Project:** Designed remote controlled car with Arduino, breadboard, C code, and AutoCad
- Clubs and Organizations:** Member of the Association for Computing Machinery, AI Student Collective, UC Davis Computer Science and Cybersecurity, USD Math Club, and Mathematics/Programming Tutor
- Other Projects:** (Available on my portfolio)